



SONiC HACKATHON 2026

PathTrace: Datapath Intent Tracing for SONiC

TEAM	PathTrace
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Problem statement



A SONiC config can be accepted, yet silently never reach the ASIC. Nothing errors, and the CLI and CONFIG_DB still show the object as present.



Example: orchagent or syncd gets a SAI error while programming, so the config never reaches the ASIC, with no user-visible failure.

1 Silent failure

The object looks configured everywhere an operator normally checks, so there is no error to start from and traffic is simply wrong.

2 Manual hunt

20–40 min per incident.

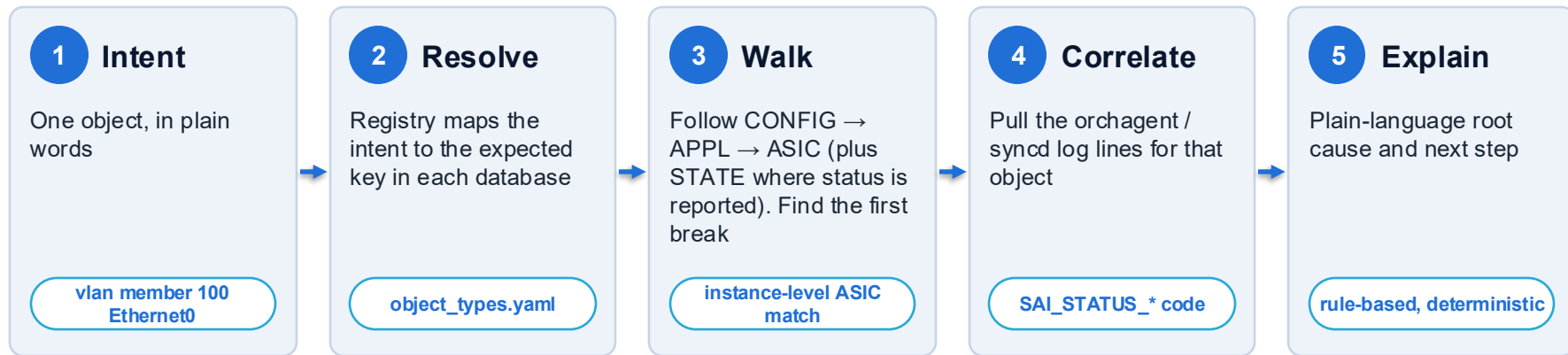
Hand-walk four Redis databases across containers, then grep orchagent and syncd logs and match the SAI error to the object.

3 Tooling gap

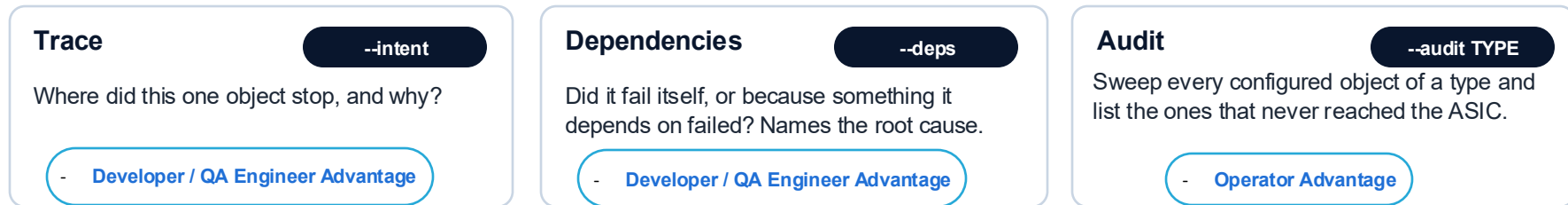
YANG and config validate check that configuration is well-formed, not whether it propagated to ASIC_DB and the silicon.

PathTrace's answer: one command traces the object across the pipeline, pinpoints exactly where it broke, and explains why using the SAI error from syslog.

Solution approach: one intent in, root cause out

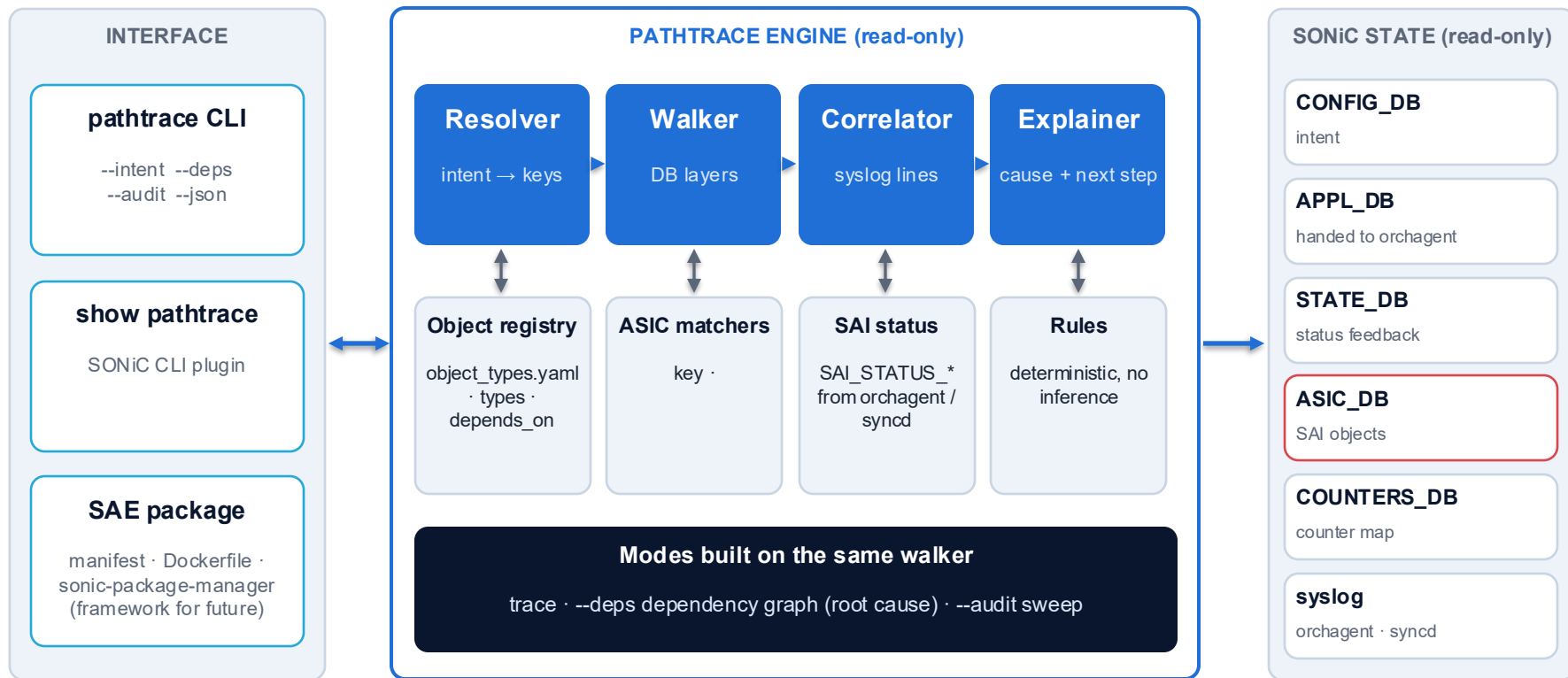


THREE WAYS TO USE IT



Read-only · reads existing Redis state and syslog only · no new SAI objects · runs on SONiC VS and hardware · deterministic (AI explanation is a future option)

Architecture



What's new for SONiC

- A read-only tool that localizes where a config stopped between CONFIG_DB and silicon
- Correlated SAI / syslog evidence for the root cause
- Instance-level matching, --deps root-cause tracing and --audit sweeps
- Declarative registry, extensible by YAML · SONiC Application Extension with native show CLI

LINKS

Repo

[\[https://github.com/palcnetworks/Path-Trace\]](https://github.com/palcnetworks/Path-Trace)

Design doc

<https://github.com/palcnetworks/Path-Trace/blob/main/pathtrace/docs/pathtrace-hld.md>

Before the hackathon: concept & design reference documents. **Built this week:** tracer, instance-level matching, --deps, --audit, SAE package.

STATUS

Completed

Tracer, instance-level matching, --deps, --audit and SAE packaging are implemented.

Future extensions feature validators on the same engine · suitable for Operator's usage. AI inferencing.